Supplementary Methods: The Role of Estradiol in Oxidative Stress in Health and Cystathionine β-Synthase Deficiency: A Mechanistic Model

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1 Mathematical Model

The model code file (as sbml file) is supplied in Data S2.

In specifying the differential equations, we use lower case letters and simple abbreviations for the variables (substrates); these abbreviations are indicated in Table 1, below. Velocities are always indicated by V_{X} where the subscript X gives the acronym of the enzyme that catalyzes that particular velocity. Each velocity depends on the current values of substrates.

Table A, Variable names and usual acronyms

Variable	Usual Acronym
Hcy	homoCysteine
Cyst	Cystathionine
GlutCys	Glutamyl-Cysteine
cGSH	cytsolic glutathione
cGSSG	cytosolic glutathione disulfide
bGSH	blood glutathion
bCys	blood Cysteine
cH_2O_2	cytosolic hydrogen peroxide
mH_2O_2	mitochondrial hydrogen peroxide
mGSH	mitochondrial glutathione
mGSSG	mitochondrial glutathione disulfide
$cTrx_{red}$	cytosolic reduced thioredoxin
$cTrx_{ox}$	cytosolic oxidized thioredoxin
$mTrx_{red}$	mitochondrial reduced thioredoxin
$mTrx_{ox}$	mitochondrial oxidized thioredoxin
E2	estradiol
P4	progesterone

Some of the reactions depend on the concentrations of other substrates that are not variable (in the model) and are assumed to be constant. These are given in Table B.

Table B. Constant concentrations (μM) and inputs (μM/hr) in the model

Abbreviation	Value	Name
c(m)NADPH	50	cytosolic (mitochondrial) nicotinamide adenine dinucleotide phosphate
METIN	100	methionine input
CYSIN	100	blood cysteine input
cGlu	3219	cytosolic glutamate
cGly	1850	cytosolic glycine

The 17 differential equations are simply mass balance equations that say that the rate of change of the concentration of a substrate is the sum of the velocities of the reactions that make the substrate minus the sum of the reactions that use the substrate. It is assumed that the mitochondria occupy 15% of the volume of the cell [1]. The full differential equations follow:

$$
\begin{aligned}
\frac{d}{dt}(Hcy) &= METIN(t) - V_{\mathrm{CBS}}(Hcy, E2, cGSH) - (3)(Hcy) \\
\frac{d}{dt}(Cyst) &= V_{\mathrm{CBS}}(Hcy, E2, cGSH) - V_{\mathrm{CTGL}}(Cyst) \\
\frac{d}{dt}(cCys) &= V_{\mathrm{CTGL}}(Cyst) - V_{\mathrm{GCL}}(cCys, cGlu, cGSH, GlutCys, cH_2O_2, E2) + V_{\mathrm{bCYSc}}(bCys, cCys) \\
&\quad -(0.35)\frac{cCys^2}{200} \\
\frac{d}{dt}(GlutCys) &= V_{\mathrm{GCL}}(cCys, cGlu, cGSH, GlutCys, cH_2O_2, E2) - V_{\mathrm{GS}}(cGly, GlutCys, cGSH) \\
\frac{d}{dt}(cGSH) &= V_{\mathrm{GS}}(cGly, GlutCys, cGSH) - V_{\mathrm{cGSHHb}}(cGSH) - V_{\mathrm{cGSHLb}}(cGSH) \\
&\quad -2V_{\mathrm{GPX_{cyto}}}(cGSH, cH_2O_2, E2, P4) + 2V_{\mathrm{GR_{cyto}}}(cGSSG, cNADPH) \\
&\quad -(0.002)cGSH - V_{\mathrm{cGSHm}}(cGSH) + \frac{.15}{.85}V_{\mathrm{mGSHc}}(mGSH) \\
\frac{d}{dt}(cGSSG) &= V_{\mathrm{GPX_{cyto}}}(cGSH, cH_2O_2, E2, P4) - V_{\mathrm{GR_{cyto}}}(cGSSG, cNADPH) \\
&\quad -V_{\mathrm{cGSSGHb}}(cGSSG) - V_{\mathrm{cGSSGLb}}(cGSSG) - (.1)cGSSG \\
\frac{d}{dt}(bGSH) &= V_{\mathrm{cGSHHb}}(cGSH) + V_{\mathrm{cGSHLb}}(cGSH) - (0.7)bGSH - (90)bGSH \\
\frac{d}{dt}(bGSSG) &= V_{\mathrm{cGSSGHb}}(cGSSG) + V_{\mathrm{cGSSGLb}}(cGSSG) - (7.5)bGSSG - (67.5)bGSSG \\
\frac{d}{dt}(bCys) &= (90)bGSH + (2)(67.5)bGSSG + CYSIN - V_{\mathrm{bCYSc}}(bCys, cCys) - (0.35)bCys \\
\frac{d}{dt}(cH_2O_2) &= V_{\mathrm{H2O2prod_{cyto}}}(t) + \frac{.15}{.85}k_{\mathrm{h2o2Transport}}(mH_2O_2) - k_{\mathrm{h2o2Transport}}(cH_2O_2) \\
&\quad -V_{\mathrm{GPX_{cyto}}}(cGSH, cH_2O_2, E2, P4) - V_{\mathrm{CAT_{cyto}}}(cH_2O_2) - V_{\mathrm{TPX_{cyto}}}(ctrx_{\mathrm{red}}, cH_2O_2) \\
\frac{d}{dt}(mH_2O_2) &= V_{\mathrm{H2O2prod_{mito}}}(t) - k_{\mathrm{h2o2Transport}}(mH_2O_2) + \frac{0.85}{0.15}k_{\mathrm{h2o2Transport}}(cH_2O_2) \\
&\quad -V_{\mathrm{GPX_{mito}}}(mGSH, mH_2O_2, E2, P4) - V_{\mathrm{CAT_{mito}}}(mH_2O_2) \\
&\quad -V_{\mathrm{TPX_{mito}}}(mtrx_{\mathrm{red}}, mH_2O_2) \\
\frac{d}{dt}(mGSH) &= -2V_{\mathrm{GPX_{mito}}}(mGSH, mH_2O_2, E2, P4) + 2V_{\mathrm{GR_{mito}}}(mGSSG, mNADPH) \\
&\quad -V_{\mathrm{mGSHc}}(mGSH) + \frac{.85}{.15}V_{\mathrm{cGSHm}}(cGSH) \\
\frac{d}{dt}(mGSSG) &= V_{\mathrm{GPX_{mito}}}(mGSH, mH_2O_2, E2, P4) - V_{\mathrm{GR_{mito}}}(mGSSG, mNADPH) - (1)(mGSSG) \\
\frac{d}{dt}(ctrx_{\mathrm{red}}) &= V_{\mathrm{TRXsyn_{cyto}}}(t) - V_{\mathrm{TPX_{cyto}}}(ctrx_{\mathrm{red}}, cH_2O_2) + V_{\mathrm{TR_{cyto}}}(ctrx_{\mathrm{ox}}, cNADPH) \\
&\quad -(0.08)(ctrx_{\mathrm{red}}) \\
\frac{d}{dt}(ctrx_{\mathrm{ox}}) &= V_{\mathrm{TPX_{cyto}}}(ctrx_{\mathrm{red}}, cH_2O_2) - V_{\mathrm{TR_{cyto}}}(ctrx_{\mathrm{ox}}, cNADPH) - (0.05)(ctrx_{\mathrm{ox}}) \\
\frac{d}{dt}(mtrx_{\mathrm{red}}) &= V_{\mathrm{TRXsyn_{mito}}}(t) - V_{\mathrm{TPX_{mito}}}(mtrx_{\mathrm{red}}, mH_2O_2) + V_{\mathrm{TR_{mito}}}(mtrx_{\mathrm{ox}}, mNADPH) \\
&\quad -(0.08)(mtrx_{\mathrm{red}})
\end{aligned}
$$

$$\frac{d}{dt}(mtrx_{\text{ox}}) = V_{\text{TPX}_{\text{mito}}}(mtrx_{\text{red}}, mH_2O_2) - V_{\text{TR}_{\text{mito}}}(mtrx_{\text{ox}}, mNADPH) - (0.05)(mtrx_{\text{ox}})$$

The details of the biochemistry and the biology are in the functional forms that show how each of the velocities depends on the current values of the variables that influence it. Many reactions have Michaelis-Menten kinetics in one of the following standard forms:

$$V = \frac{V_{max}[S]}{K_m + [S]}, \qquad V = \frac{V_{max}[S_1][S_2]}{(K_{S_1} + [S_1])(K_{S_2} + [S_2])}$$

$$V = \frac{V^f_{max}[S_1][S_2]}{(K_{S_1} + [S_1])(K_{S_2} + [S_2])} - \frac{V^b_{max}[P_1][P_2]}{(K_{P_1} + [P_1])(K_{P_2} + [P_2])}$$

for unidirectional, one substrate, unidirectional, two substrates, and bidirectional, two substrates, two products, respectively. For these reactions, Table C lists the K_m and V_{max} values. In general, we take K_m values from the literature. V_{max} values are extremely variable because they depend on enzyme expressions levels that vary in time and therefore experimental measurements *in vivo* are difficult and unreliable. We usually adjust the V_{max} values so as to obtain the typical substrate concentration values that we find in the literature. Parameters have sometimes been chosen by comparing model outputs in various circumstances to qualitative and quantitative experimental data.

Table C. Model kinetic parameters (time in hrs, concentration in μM)

	Parameter	Literature	Model	Reference
GR_{cyto}				
	$K_{m,cGSSG}$	72-107	107	[2, 3]
	$K_{m,cNADPH}$	10.4	10.4	[3]
	V_{max}		7000	
GR_{mito}				
	$K_{m,cGSSG}$	72-107	107	[2, 3]
	$K_{m,mNADPH}$	10.4	10.4	[3]
	V_{max}		5300	
TR_{cyto}				
	$K_{m,trx_{ox}}$	2.5	2.5	[4]
	$K_{m,cNADPH}$	6	6	[4]
	V_{max}		40000	
TR_{mito}				
	$K_{m,trx_{ox}}$	2.5	2.5	[4]
	$K_{m,mNADPH}$	6	6	[4]
	V_{max}		50000	
CAT_{mito}				
	K_{m,cH_2O_2}	32300	32300	[1]
	V_{max}		5.5e10	

Reactions with nonstandard kinetics follow.

CBS. The velocity of the CBS reaction is given by

$$V_{\text{CBS}}(Hcy, E2, cGSH) = \frac{V_{max}(Hcy)}{K_m + Hcy}\left(1 + \frac{V_{E2}(E2 - 0.09)}{K_{E2} + (E2 - 0.09)}\right)\left(\frac{k_{a,GSH} + cGSH}{K_{a,GSH} + sscGSH}\right) \cdot \left(\frac{k_{a,H_2O_2} + cH_2O_2}{K_{a,H_2O_2} + sscH_2O_2}\right)$$

parameter	value	reference
V_{max}	7043	
K_m	170	[5]
K_{E2}	0.67	
V_{E2}	1	
$K_{a,GSH}$	6000	
$sscGSH$	5000	
K_{a,H_2O_2}	0.01	
$sscH_2O_2$	0.01	

CBS is upregulated by estradiol [6, 7]. In [8], we chose to use a Michaelis Menten form to model CBS activity fold, dependent on E2. V_{E2} and K_{E2} were chosen so that the female-male ratio of CBS activity is 5.1/3.8 [9]. In addition, the authors in [10] found that glutathionylation of CBS increases its activity under H_2O_2-mediated oxidative stress in vitro. As the kinetics are not known yet, we added the activation of GSH and H_2O_2 using the two simple forms shown.

CTGL. The velocity of the CTGL reaction is given by

$$V_{\text{CTGL}} = \left(\frac{V_{max}(Cyst)}{K_m + Cyst}\right) \cdot \left(\frac{K_a + cH_2O_2}{K_a + sscH_2O_2}\right).$$

parameter	value	reference
V_{max}	1200	
K_m	500	[11]
K_a	0.01	[12]
$sscH_2O_2$	0.01	[1]

The activation of CTGL by H_2O_2 was not included in our previous paper [13]. However, [14] shows that the activity of CTGL increases with exogeneous and endogenous H_2O_2. As the kinetics of this activation are not clear yet, we chose the form so that it is 1 at steady state.

$\mathbf{CAT_{cyto}}$. The velocity of the CAT reaction is given by

$$V_{\text{CAT}} = \frac{V_{max}(cH_2O_2)}{K_m + cH_2O_2}\left(1 - \frac{Hcy}{IC50 + Hcy}\right).$$

parameter	value	reference
V_{max}	6.5e8	
K_m	32300	[1]
$IC50$	25	[15]

GCL. The velocity of the GCL reaction is given by

$$V_{\text{GCL}} = \frac{V_{max}((cCys)(cGlu) - \frac{GlutCys}{K_e})}{K_m^{cCys}K_m^{cGlu} + K_m^{cCys}(cGlu) + K_m^{cGlu}(cCys)\left(1 + \frac{cGSH}{K_i} + \frac{cGlu}{K_m^{cGlu}}\right) + \frac{GlutCys}{K_p} + \frac{cGSH}{K_i}} \cdot \left(\frac{sscH_2O_2 + K_a}{cH_2O_2 + K_a}\right)\left(1 + \frac{V_{E2}(E2 - 0.09)}{K_{E2} + (E2 - 0.09)}\right).$$

parameter	value	reference
V_{max}	3600	
K_m^{cCys}	100	[13]
K_m^{cGlu}	1900	[13]
K_i	8200	[13]
K_e	5597	[13]
K_p	300	[13]
K_a	0.01	[13]
$sscH_2O_2$	0.01	[1]
K_{E2}	10.8	[16]
V_{E2}	25.77	[16]

The general kinetics and parameters for GCL is taken from [13]. The addition of the E2 stimulation comes from our previous paper [16]. In that paper, we used a linear increasing function of E2. To make this model more physiological, we have changed this function to be a Michaelis-Menten function such that it saturates at high estradiol levels. We chose the parameter values that represent an 80% increase in activity between males and females as is the consensus in the literature.

GS. The velocity of the GS reaction is given by

$$V_{\text{GS}} = \frac{V_{max}((cGly)(GlutCys) - \frac{gsh}{K_e})}{K_m^{GlutCys}K_m^{cGly} + K_m^{cGly}(GlutCys) + K_m^{GlutCys}(cGly)\big(1 + \frac{cGSH}{K_i} + \frac{GlutCys}{K_m^{GlutCys}}\big) + \frac{GlutCys}{K_p}}$$

parameter	value	reference
V_{max}	5400	
K_m^{cGly}	300	[13]
$K_m^{GlutCys}$	22	[13]
K_e	5600	[13]
K_p	30	[13]

$\mathbf{GPX_{cyto}}$. The velocity of the GPX reaction is given by

$$V_{\text{GPX}} = \frac{V_{max}(cGSH)^2(cH_2O_2)}{(K_m^{gsh} + cGSH)^2(K_m^{cH_2O_2} + cH_2O_2}) \cdot \big(1 - \frac{V_{P4}(P4 - .318)}{K_{P4} + (P4 - 0.318)}\big) \cdot \big(1 + \frac{V_{E2}(E2 - 0.09)}{K_{E2} + (E2 - 0.09)}\big) \big(\frac{K_i + 1.8}{K_i + Hcy}\big).$$

parameter	value	reference
V_{max}	1.8e6	
K_m^{cGSH}	1330	[13]
$K_m^{cH_2O_2}$	6.8	[1]
V_{E2}	27	
K_{E2}	11	
V_{P4}	0.3	
K_{P4}	70	
K_i	27	[17]

The general kinetics and parameters for GPX is taken from [13]. The addition of the E2 stimulation comes from our previous paper [16]. In that paper, we used a linear increasing function of E2. To make this model more physiological, we have changed this function to be a Michaelis-Menten function such that it saturates at high estradiol levels. Multiple studies have shown that progesterone decreases GPX activity or GPX enzyme concentrations in vitro [18, 19, 20]. In [18], rat hepatic stellate cells were treated with estradiol (E2) or progesterone (P4) for 24 hours in vitro, and found that progesterone had a smaller effect on GPX activity than estradiol. For example, a 100 nM dose of estradiol increased GPX activity by 140% and a 100 nM dose of progesterone deceased the activity of GPX by 25%. Thus, in our model the strength of estradiol activation on GPX is larger than the progesterone inhibition on GPX. We chose the parameter values that represent an 80% increase in activity between males and females as is the consensus in the literature.

$\mathbf{GPX_{mito}}$. The velocity of the GPX reaction is given by

$$V_{\text{GPX}} = \frac{V_{max}(cGSH)^2(cH_2O_2)}{(K_m^{gsh}+cGSH)^2(K_m^{cH_2O_2}+cH_2O_2} \cdot \big(1-\frac{V_{P4}(P4-.318)}{K_{P4}+(P4-0.318)}\big)\cdot\big(1+\frac{V_{E2}(E2-0.09)}{K_{E2}+(E2-0.09)}\big).$$

parameter	value	reference
V_{max}	2.5e6	
K_m^{cGSH}	1330	[13]
$K_m^{cH_2O_2}$	6.8	[1]
V_{E2}	27	
K_{E2}	11	
V_{P4}	0.3	
K_{P4}	70	

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$\mathbf{TPX_{cyto}}$. The velocity of the cytosolic TPX reaction is given by

$$V_{\text{TPX}} = \frac{V_{max}(trx_{red})(cH_2O_2)}{\left(K_m^{ctrx_{red}} + ctrx_{red}\right)\left(K_m^{cH_2O_2} + cH_2O_2\right)} \cdot \left(1 - \frac{cH_2O_2}{IC50 + cH_2O_2}\right).$$

parameter	value	reference
V_{max}	3e5	
$K_m^{ctrx_{red}}$	2.4	[21]
$K_m^{cH_2O_2}$	0.7	[21]
$IC50$	0.17	[22]

The cytosolic and mitochondrial thioredoxin peroxidase can become overoxidized at high levels of H_2O_2. The kinetics for this inhibition come from [22].

$\mathbf{TPX_{mito}}$. The velocity of the cytosolic TPX reaction is given by

$$V_{\text{TPX}} = \frac{V_{max}(mtrx_{red})(mH_2O_2)}{\left(K_m^{mtrx_{red}} + mtrx_{red}\right)\left(K_m^{mH_2O_2} + mH_2O_2\right)} \cdot \left(1 - \frac{cH_2O_2}{IC50 + cH_2O_2}\right).$$

parameter	value	reference
V_{max}	8e5	
$K_m^{cH_2O_2}$	0.7	[21]
$K_m^{mtrx_{red}}$	4.3	[23]
$IC50$	1.8	[22]

The cytosolic and mitochondrial thioredoxin peroxidase can become overoxidized at high levels of H_2O_2. The kinetics for this inhibition come from [22].

Thioredoxin Synthesis and Removal. Mitochondrial and cytosolic thioredoxin are proteins trascribed and translated in cells. As the data is limited on the rate of synthesis, we take miochondrial and cytosolic thioredoxin synthesis to be 0.1 $\mu M/hr$. Multiple studies have measured the concentration of thioredoxin in cells to range from 1 to 10 μM and most of the thioredoxin pool is in its reduced form in the mitochondria and in the cytosol [24, 25, 26, 27]. Thus, in our model the cytosolic and mitochondrial reduced thioredoxin levels are around 1 μM and the oxidized forms are around 0.01 μM. The removal rates were chosen to balance the synthesis rate and maintain

these steady states. The removal rate of the reduced forms of Trx is 0.08 $1/hr$ and is 0.05 $1/hr$ for the oxidized forms.

Hydrogen Peroxide Production. In the mitochondria, H_2O_2 is produced largely from stray electrons falling off of the electron transport chain that form superoxide which is then converted to H_2O_2 via Superoxide dismutase (SOD). Cytosolic H_2O_2 is produced by the endoplasmic reticulum and peroxisomes [28]. Mitochondrial hydrogen peroxide production is taken from [1] where calculations were done to get the true rate of mitochondrial H_2O_2 production accounting for efflux based on [29] and [30]. The rate of production was calculated to be 1 nmol/min/mg protein which converts roughly to 12000 $\mu M/hr$ (assuming 1 $\mu mol/hr/mg$ protein is 1 $\mu mol/hr/5 \times 10^{-6} L$). In our model, cytosolic H_2O_2 production is taken to be 3026 $\mu M/hr$. This value was chosen to align with reports concluding that H_2O_2 production from other cellular sources contributes more to the overall liver H_2O_2 production than the mitochondria [31]. Thus, when accounting for the mitochondria and cytosol volumes, in our model the cytosol contributes to about 60% of H_2O_2 production in the liver. In addition, the production of H_2O_2 in the liver is estimated to be 50 $nmol/min/g$ tissue [28]. If the liver is assumed to be 70% water by weight, then this converts to about 4300 $\mu M/hr$ and in our model the production rate is about 4400 $\mu M/hr$ (when correcting for volumes).

Mitochondrial Transport Kinetics. GSH and H_2O_2 are the only substrates that can cross the mitochondria membranes in our model. The mitochondrial and cytosolic Trx systems are different from one another -Trx2, TPX 3, and TR 2 are in the mitochondria and Trx1, TPX2, and TR1 are in the cytosol [32]. Thus, we do not show the two systems interacting with one another through transport into/out of the mitochondria. GSSG must first be converted to GSH to be transported out. Although a mitochondrial transporter has been discovered for yeast, it remains to be proven for humans [33]. Instead, we add a small removal rate of mitochondrial GSSG to maintain balance at high levels of H_2O_2 (denoted as $(1)(mGSSG)$ in the differential equation). GSH is transported into the mitochondria via high and low affinity transporter, each following Michaelis-Menten Kinetics, while transport to the cytosol is assumed to be linear. The equation forms and constants are taken from [1].

GSH Transport

Parameter	Value
V_{cGSHm}	
High Affinity Transporter	
V_{max}	388
K_m	36.4
Low Affinity Transporter	
V_{max}	2711
K_m	9000
V_{mGSHc}	
K_{GSH}	1.03

H_2O_2 diffuses to/from the mitochondria, facilitated by aquaporins [34]. The rate was chosen to correspond with experimental data from [35], which found the half-life of H_2O_2 to be 20 seconds in the medium when incubating mitochondria with exogenous H_2O_2.

H_2O_2 Transport

Parameter	Value
$k_{H2O2Transport}$	120

Transport to/from blood. The kinetics of transport between the cytosol and the blood compartments follow [13]. For Cysteine, the general form for the kinetics of transport is taken to be

$$V = \frac{V_{max}(bCys)}{K_m + bCys} - k_{out}(cCys).$$

Cysteine Transport

Parameter	Value
V_{max}	14950
K_m	2100
K_{out}	1

The transport of Cysteine into the blood from outside is assumed to be constant at rate of $CYSIN = 105$. Also included in the differential equation for cytosolic Cysteine is the loss of cytosolic Cysteine to other metabolic processes,

$$-(0.35)\frac{cCys^2}{200}.$$

This form was chosen in [13] to account for the fact that as cytosolic Cysteine rises an increasing fraction of it is sent towards the synthesis of taurine.

High and low affinity transporters operate to export GSH and GSSG. The kinetic forms and constants follow [13].

GSH/GSSG Transport			
Parameter	Value	Parameter	Value
V_{cGSHHb}		V_{cGSHLb}	
K_m	150	K_m	3000
V_{max}	330	V_{max}	1100
		Hill	3
$V_{cGSSGHb}$		$V_{cGSSGLb}$	
K_m	1250	K_m	7100
V_{max}	40	V_{max}	4025

GSH and GSSG breakdown in the blood. GSSG and GSH breakdown into their component amino acids in the blood. Also, we assume that normally 10% of Cysteine in the blood is taken up by other cells and an additional 25% is converted to Cystine [13].

2 Model Calibration to Reproduce Experimental Results

Various aspects of H_2O_2 scavenging and its transport across the mitochondrial membrane were calibrated to align with experimental results from rat liver mitochondria. This ensured that the time dependence and kinetics of our model matched experimental observations. To reproduce the experimental conditions, we developed a mitochondrial submodel in which GSH and cytosolic H_2O_2 production, cytosolic H_2O_2 scavenging, and transport of GSH, GSSG, and Cys into/out of the cell were turned off. In addition, all plasma and cytosolic variables were set to zero except for cytosolic H_2O_2, which represents the exogenous H_2O_2 in the experimental set up.

The transport rate of H_2O_2 across the mitochondrial membrane was chosen to reproduce *in vitro* experiments from [35], in which rat liver mitochondria were incubated with exogenous H_2O_2 and the rate of disappearance was measured. In panel A of Figure 1, we overlay model results with those seen in [35]. Our model captures the change in extra-mitochondrial H_2O_2 well.

The three H_2O_2 scavenging mechanisms exist both in the mitochondria and in the cytosol of liver cells. Many studies have examined the relative contributions of different antioxidant mechanisms in erythrocytes and the mitochondria of aortic and brain cells [37, 38, 39, 40, 41, 36]. The findings vary across tissues, with no single scavenger dominating universally. Competitive kinetic analyses reveal that thioredoxin peroxidases (**TPX**) have higher reaction rates than glutathione peroxidase (**GPX**) and catalase (**CAT**) [42]. Additionally, **TPX** enzymes have lower K_m values, suggesting they dominate at low steady-state levels of H_2O_2. In contrast, it is reasonable to conclude **GPX** is more effective at intermediate concentrations, while **CAT**, with its very high K_m, becomes the primary scavenger at elevated H_2O_2 levels. However, studies done in liver mitochondria of rats tell a different story. When 3 μM of H_2O_2 was given exogenously to rat liver mitochondria, there was a

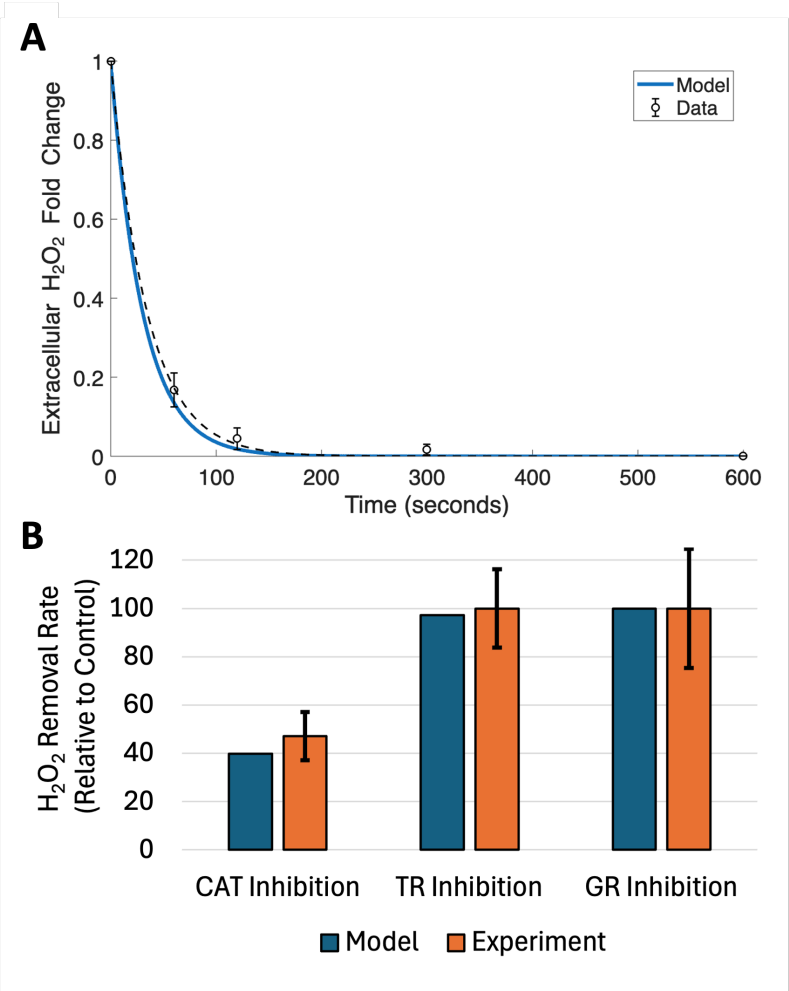

Figure 1: **Model Recapitulates Experimental Results.** Panel A shows the decay of 1000 μM H_2O_2 when added exogenously to our mitochondria submodel (blue line) and to rat liver mitochondria as in [35] (circle). In panel B, the H_2O_2 removal rate, relative to control, is shown when 3 μM H_2O_2 is added exogenously to our mitochondrial submodel (blue bars) and to rat liver mitochondria [36] (orange bars). The error bars represent standard errors in the experiments. To create the experimental set up, our mitochondria submodel turns off GSH production, the cytosolic H_2O_2 scavenging mechanisms, and transport of GSH, GSSG, and Cys into/out of cell. In addition, all plasma and cytosolic variables, aside from cytosolic H_2O_2 are set to zero. The cytosolic H_2O_2 represents the exogenous H_2O_2 in the experimental set up.

50% decrease in H_2O_2 removal rates when **CAT** was inhibited. However, no significant difference in removal rates was seen for either thioredoxin reductase (**TR**) or glutathione reductase (**GR**) inhibitors [36]. Further, another study in rat liver mitochondria found that when thioredoxin reductase (**TR**) was inhibited, H_2O_2 production did not change significantly when endogenous respiratory substrates were used [43]. Thus, in liver mitochondria, it seems to be the case that at steady state levels of H_2O_2, **CAT** is the dominating H_2O_2 scavenger. We chose the V_{max} values

of mitochondrial **CAT**, **TPX**, and **GPX**, so that at steady state levels of H_2O_2, the relative contribution of **CAT**-mediated H_2O_2 scavenging is 80%, while **TPX** and **GPX** each contribute 10%.

Experimental studies assessing the relative importance of different H_2O_2 removal mechanisms often do so by selectively inhibiting specific antioxidant enzymes [37, 44, 43, 36, 35]. In Table 2 of [36], the authors report removal rates of exogenous H_2O_2 when incubated with rat liver mitochondria. Panel B in Figure 1 illustrates how the H_2O_2 removal rates change under the same inhibition scenarios used in [36]. Consistent with their findings, our results show that inhibition of **GR** and **TR** does not alter H_2O_2 removal rates, whereas the inhibition of **CAT** reduces removal rates by approximately 50%.

The relative contributions of cytosolic H_2O_2 scavenging in the liver are less well established, and little direct data are available. Thus, we chose the V_{max} values of cytosolic **CAT**, **TPX**, and **GPX**, so that at steady state levels of H_2O_2, **GPX** accounts for approximately 50% of H_2O_2 scavenging, **TPX** for 40% and **CAT** for 10%. This follows [45], which found that catalase played a minor role in H_2O_2 unless H_2O_2 production is high.

3 Sensitivity Analysis and Virtual Population Generation

In order to create a virtual population to analyze the variability of the substrate responses to estradiol at steady state and during the menstrual cycle, we first performed a Sobol sensitivity analysis to determine which parameters were the most significant in driving cytosolic H_2O_2 levels. The following parameters were not included in the analysis and were held constant:

- P4 and E2
- P4 and E2 stimulation parameters
- METIN
- CYSIN
- NADPH and kinetic parameters
- cGly
- cGlu

All other parameters were sampled within 50% and 150% of the nominal values. We used log-uniform distributions for each parameter, and used the Sobol sequence sampling scheme. Matlab's *Simbiology* sbiosobol function was used for this step. Parameter sampling continued until the set

of top 20 parameters ranked by Total-Order Sobol Index stabilized. Please see Table S1 for the exact ranges.

We then selected the top 15 parameters to construct a virtual population of $N = 20,000$ individuals using the plausible patient generation framework described in [46]. For this, we supplied plausible parameter and species ranges. The parameters were sampled between 50% and 150% of nominal values. The parameter and species ranges are provided in Table S2 and the final parameter and species values for the virtual population are provided in Table S3.

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